

Homework #1 GPS Collection Planning using Navmatrix GPS Mission Planning

This assignment includes two parts:

- 1) You will follow the instruction to plan the mission for our class project
- 2) You will use what you learned in this instruction to plan the mission for your own project

Part I

This assignment will require you to use the Navmatrix GPS Mission Planning website:

<http://gnssmissionplanning.com/>.



Click on 'Begin Planning', and navigate to the Settings Tab.



You will plan for a GPS collection on 10/04/2022 at the University of the Central Arkansas. You will need to fill the settings form. Please read carefully.

- **Search Address:** set to 'University of central Arkansas', then click on search icon . You will notice, the map will zoom to the UCA. The latitude and longitude information will be automatically filled. Note: If you know the latitude and longitude information, you can fill the latitude and longitude information directly, in that case, you do not need to search Address. For example, you can directly enter Latitude 39.5° N (entered as 39.5) and Longitude 77.3° W (entered as -77.3).

Search Address

University of central Arkansas

Search Results

University of Central Arkansas, 201. Donaghey Avenue, Conway, Faulkner County

Latitude

35.07764105

Longitude

-92.45799369818861

Height

0

meters (m)

Cutoff

10

degrees (°)

Date

10/04/2022

From / For

12:00 PM

8h

Time zone

(UTC-06:00) Central Time (US & Canada)

-Height: set to 0 meters

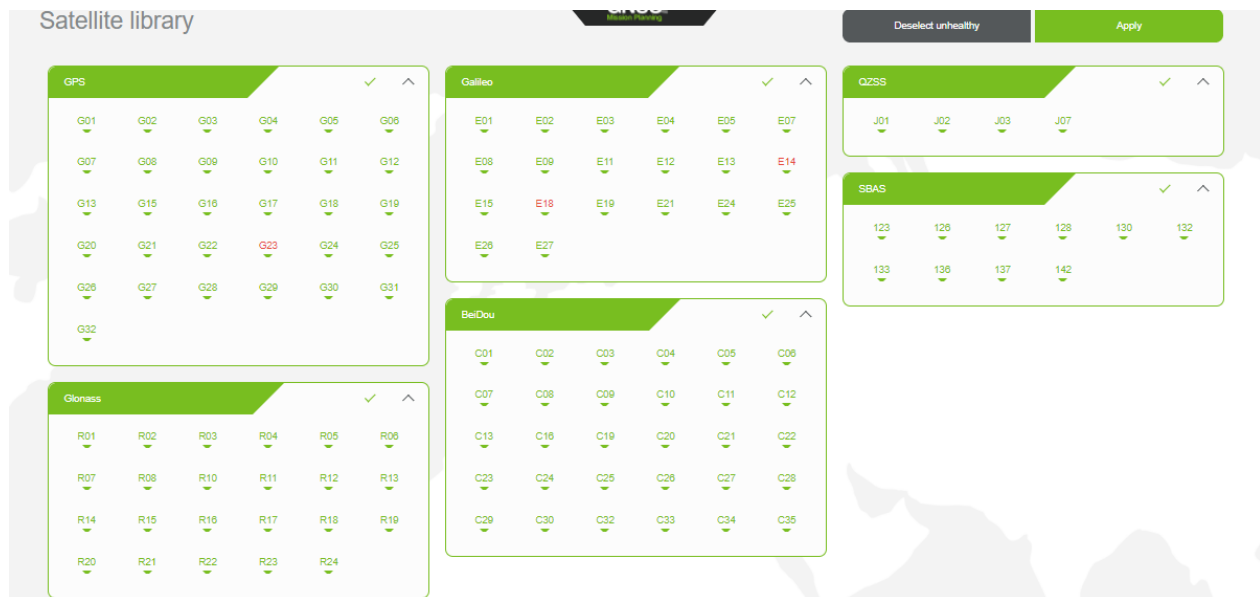
-Cutoff: set to 10 degree.

-Date: set to 10/04/2022

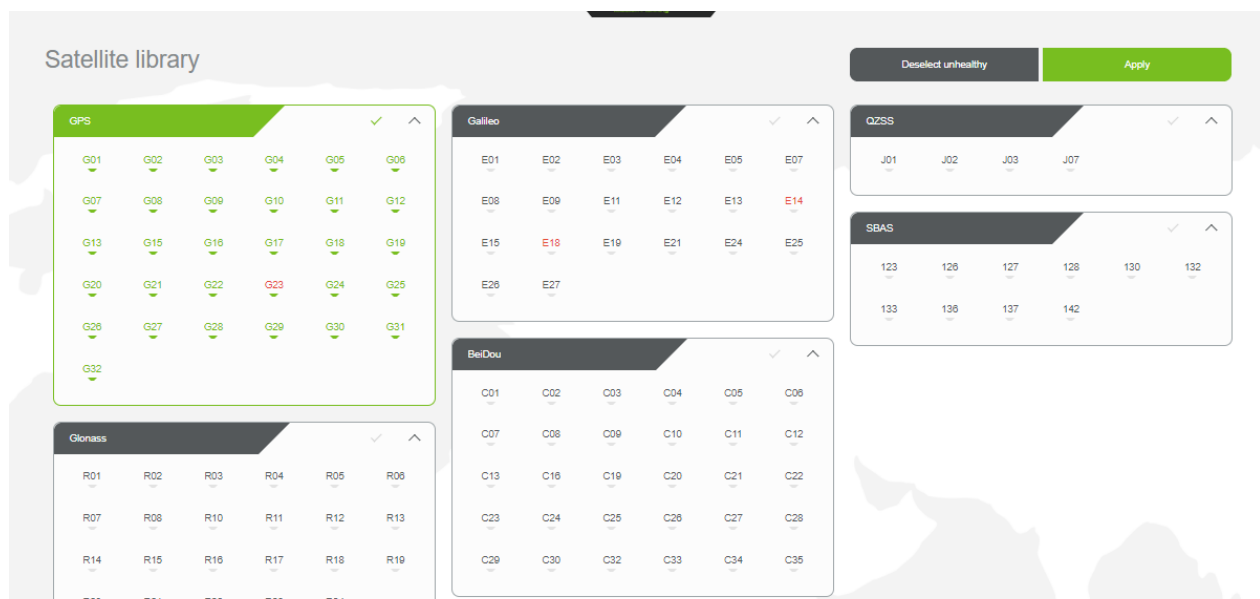
-From/For: 12:00 PM for 6 hours. Note, here assume the data collection time period is from 12: 00 pm to 6: 00 pm.

-Time Zone: Set 'UTC-6:00 Central Time (US & Canada)'.

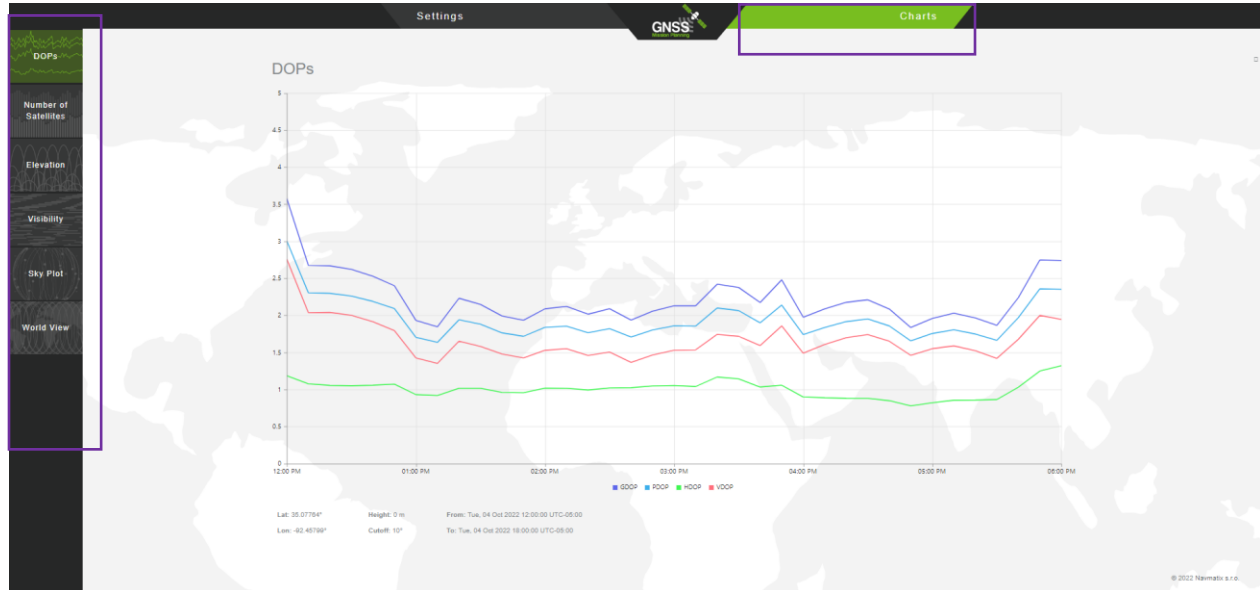
After finish setting, click on 'Apply'. You will see available GNSSs. **Note**, you may need to wait for couple minutes.



Turn off all of the constellations excepting GPS by clicking on Galileo, QZSS, SBAS, GLONASS, and BeiDou.



Click on 'Apply'. It will start to calculate. Note, you may need to wait for couple minutes. When it finished, the 'Charts' will be activated. You can navigate to the left panel to read different indexes.



Note, your time may not be shown as 12:00 pm to 6:00 pm as mine. If you select the correct time zone (i.e., UTC-6:00 Central Time (US & Canada)), it should be right.

With the indexes, please answer the following questions (**Be careful, do not click on World View at this time, otherwise your results will be lost!**).

1. During which time period (roughly) is relatively better to collect the data? Why? (Hint, the lower the GDOP/PDOP/HDOP/VDOP, the higher the accuracy) (5 points)
2. What is the maximum number of GPS satellites that are predicted to be detected during this time period? (5 Points) (Hint, please refer to 'Number of Satellites' panel)
3. On this time period and at this location, when will satellite G01 be observable? (5 Points) (Hint, please refer to 'Visibility' panel)

Part II

Now you will do your own mission planning. You could need to select different time or location as that in Part I.

- 1) Copy the Setting form to your word document. (Note, you can keep height and cutoff as Part I) (5 points)
- 2) Copy the DOPs graph to your document (5 points)
- 3) Copy the number of Satellites to your document (5 points)
- 4) Copy the Visibility to your document (5 points)
- 5) Copy the Sky plot at end time to your document (Hint, you need to scroll the time icon to most right) (5 points)
- 6) When is the best time for you to collect your project data? (10 points)

Homework

Please copy these questions with your answers to a word document and submit it through the Blackboard.